

Darwix Assignment — Submission Overview

AI-powered health insurance voice agent with a grounded knowledge base, retrieval pipeline, qualification workflow, CRM integration, and neural voice responses.

5/5 Retrieval scenarios validated	5 End-to-end call scenarios	HIGH Retrieval confidence across benchmark	CRM Lead & quote automation
--------------------------------------	--------------------------------	---	--------------------------------

Solution Highlights

- **Knowledge ingestion & cleaning:** Structured extraction from HTML, policy text, FAQ/objection content, underwriting JSON, and customer logs, with boilerplate removal, PII redaction, terminology normalization, and duplicate detection.
- **Hybrid retrieval:** Combines BM25 lexical search with TF-IDF semantic similarity and Reciprocal Rank Fusion (RRF), producing ranked results with confidence estimation.
- **Grounded responses:** Responses are tied to retrieved knowledge-base context and include source/record provenance for traceability.
- **Conversational qualification:** Maintains caller state including name, age, pre-existing conditions, preferred plan, qualification status, and escalation state.
- **Business workflow:** Calculates preliminary premiums, creates CRM leads, records qualification status, and retains the citations used for each interaction.
- **Voice experience:** FastAPI-powered calling interface with neural speech synthesis, cached audio generation, session handling, and APIs for calls, knowledge search, CRM leads, and health checks.

Validation & Demonstrated Scenarios

Area	Demonstrated capability
Customer qualification	Captures applicant details and produces a plan quote with business-action output.
Objection handling	Uses grounded plan, pricing, coverage, and pre-existing-condition information.
Eligibility & escalation	Identifies senior-age cases and routes them to the appropriate underwriting flow.
Safe handling	Provides controlled responses for unsupported/out-of-scope requests.
Human assistance	Supports explicit escalation to a supervisor/human assistance workflow.

Technical Stack

Python · FastAPI · Pydantic · BM25 · TF-IDF · NumPy · scikit-learn · BeautifulSoup · Gemini/OpenAI integrations · Edge-TTS · JSON persistence

Repository

Complete implementation, test runs, generated audio artifacts, retrieval benchmark, and documentation are included in the submitted GitHub repository.